Liquid intrusion and evaporation in low-pressure cavities: numerical investigation with STAR-CCM+

Einströmung und Verdampfung einer Flüssigkeit unter Niedrigdruckbedingungen: numerische Simulationen mit STAR-CCM+

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Masterarbeit

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1 Numerical Methods

This chapter contains an overview of the governing equations used in Star-CCM+ for the modelling of multiphase flows with a focus on eulerian multiphase (EMP) flow and the volume of fluid (VOF) method. Turbulence and the Shear Stress Transport (SST) k-ω model is briefly discussed. The physical processes of evaporation and vaporisation are described. The ability to model multiple flow regimes, capture interfaces, and the phase interaction closures used in Star-CCM+ are discussed. The chapter concludes with a description of the numerical methods used in Star-CCM+ along with the apparent limitations [better word] of the commercial software in being able to accurately model the problems under consideration.

1.1 Governing Equations

[this section is somewhat rough]

Multiphase flow is modelled in Star-CCM+ using various frameworks. The two-fluid - eulerian multiphase - framework solves a set of six equations to model two-phase flow. A set of transport equations for mass, momentum, and energy are solved for all the phases, which are assumed to be interpenetrating continua that share a common pressure field. The local instantaneous conservation equations are assumed to be euler averaged and contain closures based on interfacial terms that define the influences of the phases upon each other.

Although used primarily for modelling dispersed flows, the EMP framework is capable of resolving both dispersed and separated flow regimes.

The implemented volume of fluid method in Star-CCM+ assumes sufficient mesh resolution in order to resolve the interface and model separated flows. The interface capturing method solves the volume fraction transport equation using the High-Resolution Interface Capturing (HRIC) scheme to maintain a sharp interface between the phases.

Based on the local volume fraction of the phase i, $\alpha_i = V_i/V$, where in each cell $\sum_i \alpha_i = 1$. The material properties in each cell and other flow variables are assumed to be a mixture of the properties of the phases as follows:

$$\text{density, } \rho = \textstyle\sum_i \alpha_i \rho_i, \tag{1.1a}$$

$$\text{viscosity, } \mu = \textstyle\sum_i \alpha_i \mu_i, \tag{1.1b}$$

$$\text{velocity, } \underline{\mathbf{v}} = \textstyle\sum_i \alpha_i \rho_i \underline{\mathbf{v}}_i / \rho, \tag{1.1c}$$

$$\text{specific heat, } C_p = \textstyle\sum_i \alpha_i C_{p,i} \rho_i / \rho, \tag{1.1d}$$

$$\text{thermal conductivity, } k = \textstyle\sum_i \alpha_i k_i, \tag{1.1e}$$

$$\text{enthalpy, } \mathrm{H} = \textstyle\sum_i \alpha_i \rho_i \mathrm{H}_i / \rho + |\underline{\mathbf{v}}|^2/2. \tag{1.1f}$$

where ρ_i is the density, μ_i is the dynamic viscosity, $\underline{\mathbf{v}}_i$ is the velocity, $C_{p,i}$ is the specific heat, k is the thermal conductivity, and H_i is the enthalpy of the phase i.

The diffusion velocity of phase i, $\underline{\mathbf{v}}_{d,i}$, is defined as the difference between the velocity of the phase and the mixture such that

$$\underline{\mathbf{v}}_{d,i} = \underline{\mathbf{v}}_i - \underline{\mathbf{v}}.$$

A set of volume fraction transport and mass, momentum, and energy conservation equations are solved for each phase. In cases of two-phase flow, Star-CCM+ solves the volume fraction transport equation only for the first phase.

1.1.1 Eulerian Multiphase Flow

The euler averaged mass conservation equation used in the eulerian multiphase formulation is shown in 1.2. The right hand side of the equation contains the user defined mass source term S_i^α and the interfacial mass source Γ_k. The interfacial mass source is the net mass transfer per unit volume from phase j to phase i, m_{ij}, and from phase i to phase j, m_{ji}.

$$\frac{\partial}{\partial t}\int_V \alpha_i\rho_i \,\mathrm{d}V + \int_{\partial V} \alpha_i\rho_i\underline{\mathbf{v}}_i\cdot \mathrm{d}\underline{\mathbf{a}} = \Gamma_i + \int_V \mathrm{S}_i^\alpha \,\mathrm{d}V. \tag{1.2}$$

$$\Gamma_i = \int_V \sum_{j\neq i}(m_{ij} - m_{ji})\,\mathrm{d}V. \tag{1.3}$$

The governing momentum conservation equation implemented in the eulerian multiphase formulation is given in Eqn. (1.4). The right hand side of the equation consists the various body and surface forces acting on the fluid volume element. The pressure field, p, is assumed to be shared by both the phases. The vector $\underline{\mathbf{g}}$ denotes the gravitational acceleration and the laminar and turbulent stresses are given by $\underline{\underline{\mathbf{T}}}_i$ and $\underline{\underline{\mathbf{T}}}_i^t$ respectively. An additional user defined momentum source term can be appended to the equation using $\underline{\mathbf{S}}_i^\alpha$. Internal forces such as user defined potential forces or solid pressure forces between particles can be accounted for using the $(\underline{\mathbf{F}}_{\mathrm{int}})_i$ term.

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \alpha_i\rho_i\underline{\mathbf{v}}_i \,\mathrm{d}V + \int_{\partial V} \alpha_i\rho_i\underline{\mathbf{v}}_i\otimes\underline{\mathbf{v}}_i\cdot \mathrm{d}\underline{\mathbf{a}} = -\int_V \alpha_i\nabla p\,\mathrm{d}V + \int_V \alpha_i\rho_i\underline{\mathbf{g}}\,\mathrm{d}V + \int_{\partial V}\left[\alpha_i\left(\underline{\underline{\mathbf{T}}}_i + \underline{\underline{\mathbf{T}}}_i^t\right)\right]\cdot \mathrm{d}\underline{\mathbf{a}} \\ + \int_V (\underline{\mathbf{F}}_{\mathrm{int}})_i\,\mathrm{d}V + \int_V \underline{\mathbf{S}}_i^\alpha\,\mathrm{d}V + \mathrm{MomTerm}.\end{aligned} \tag{1.4}$$

The term Γ groups the interphase, interfacial momentum transfer terms together The interphase and interfacial momentum transfer is accounted through

$$MomTerm = \int_V \underline{\mathbf{M}}_i\,\mathrm{d}V + \int_V \sum_{j\neq i}(m_{ij}\underline{\mathbf{v}}_j - m_{ji}\underline{\mathbf{v}}_i)\,\mathrm{d}V \tag{1.5}$$

The momentum transfer, that satisfies the equation $\sum_i \underline{\mathbf{M}}_i = 0$, contains contributions from the drag, lift, virtual mass, turbulent dispersion, and wall lubrication force respectively. These forces are defined as force per cell volume and it follows that $\underline{\mathbf{F}}_{ij} = -\underline{\mathbf{F}}_{ji}$.

$$\underline{\mathbf{M}}_i = \sum_{j\neq i}\left(\underline{\mathbf{F}}_{ij}^{\mathrm{D}} + \underline{\mathbf{F}}_{ij}^{\mathrm{L}} + \underline{\mathbf{F}}_{ij}^{\mathrm{VM}} + \underline{\mathbf{F}}_{ij}^{\mathrm{TD}} + \underline{\mathbf{F}}_{ij}^{\mathrm{WL}}\right). \tag{1.6}$$

The conservation of energy as solved in the eulerian multiphase formulation is shown in 1.7 Although written here for total energy of the phase, E_i, the equation can be either solved for the total enthalpy, H_i, or the temperature T_i.

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \alpha_i\rho_i E_i\,\mathrm{d}V + \int_{\partial V}\alpha_i\rho_i\underbrace{[h_i^{\mathrm{REF}} + \int_{T_i^{\mathrm{REF}}}^{T_i} C_{p,i}(T')\,\mathrm{d}T' + \frac{1}{2}\underline{\mathbf{v}}_i\cdot\underline{\mathbf{v}}_i]}_{\text{total enthalpy, } H_i}\underline{\mathbf{v}}_i\cdot \mathrm{d}\underline{\mathbf{a}} = \int_{\partial V}\alpha_i(k_{\mathrm{eff}})_i\nabla T_i\cdot \mathrm{d}\underline{\mathbf{a}} + \int_{\partial V}\underline{\underline{\mathbf{T}}}_i\underline{\mathbf{v}}_i\cdot \mathrm{d}\underline{\mathbf{a}} \\ + \int_V \underline{\mathbf{f}_{\mathbf{b}}}_i\cdot\underline{\mathbf{v}}_i\,\mathrm{d}V + \int_V \mathrm{S}_{u,i}\,\mathrm{d}V + EnTerm.\end{aligned} \tag{1.7}$$

The total energy can be related to the total enthalpy of the phase using $\mathrm{E}_i = \mathrm{H}_i - p/\rho_i$. The right hand side of the equation consists of the viscous stress tensor $\underline{\underline{\mathbf{T}}}_i$, the body force vetor $\underline{\mathbf{fb}}_i$, and the effective thermal conductivity $(k_{\mathrm{eff}})_i$. The effective thermal conductivity of the phase i is defined using laminar and turbulent contributions as

$$(k_{\mathrm{eff}})_i = k_i + \frac{\mu_{t,i} C_{p,i}}{\sigma_{t,i}},$$

where k_i is the thermal conductivity, $\mu_{t,i}$ the turbulent viscosity, $C_{p,i}$ the specific heat, and $\sigma_{t,i}$ the turbulent thermal diffusion Prandtl number of the phase.

Along with a user-defined energy source term $\mathrm{S}_{u,i}$, the equation contains the interfacial and interphase energy terms that are gathered together and denoted using $EnTerm$.

$$EnTerm = \int_V \sum_{j \neq i} Q_{ij} \,\mathrm{d}V + \int_V \sum_{ij} Q_i^{(ij)} \,\mathrm{d}V + \int_V \sum_{j \neq i} (m_{ij} - m_{ji}) h_i(T_{ij}) \,\mathrm{d}V \tag{1.8}$$

Q_{ij} is the heat transfer from phase i to phase j due to temperature difference between the phases, $Q_i^{(ij)}$ is the heat transfer between the phase i and the interface which accounts for phenomenon such as evaporation and boiling, $h_i(T_{ij})$ is the enthalpy of the phase i evaluated at the temperature T_{ij} of the interface. The interphase energy terms are related as follows:

- for cases where heat is exchanged only through thermal diffusion, the interphase heat transfer rates satisfy $Q_{ij} = -Q_{ji}$
- for cases where heat and mass are transferred only through phase change, the interphase heat and mass transfer rates satisfy $Q_i^{(ij)} + Q_j^{(ij)} + \left(m_{ij} - m_{ji}\right)\left(h_j(T_{ij}) - h_i(T_{ij})\right) = 0$.

1.1.2 Volume of Fluid Method

The distribution of the phase i is driven by the volume fraction transport.

$$\frac{\partial}{\partial t} \int_V \alpha_i \,\mathrm{d}V + \int_{\partial V} \alpha_i \underline{\mathbf{v}}_i \cdot \mathrm{d}\underline{\mathbf{a}} = \int_V \left(\mathrm{S}_i^\alpha - \frac{\alpha_i}{\rho_i} \frac{D\rho_i}{Dt} \right) \mathrm{d}V - \int_V \frac{1}{\rho_i} \nabla (\alpha_i \rho_i \underline{\mathbf{v}}_{d,i}) \,\mathrm{d}V \tag{1.9}$$

where $\underline{\mathbf{a}}$ is the surface area vector, $\underline{\mathbf{v}}$ is the mixture velocity, $\underline{\mathbf{v}}_{d,i}$ is the slip velocity, and S_i^α is the user defined phase source term for the phase i.

The **mass conservation** equation solved is

$$\frac{\partial}{\partial t} \int_V \rho \,\mathrm{d}V + \int_{\partial V} \rho \underline{\mathbf{v}} \cdot \mathrm{d}\underline{\mathbf{a}} = \int_V \sum_i \mathrm{S}_i^\alpha \rho_i \,\mathrm{d}V \tag{1.10}$$

The **momentum conservation** equation solved is

$$\begin{aligned} \frac{\partial}{\partial t} \int_V \rho \underline{\mathbf{v}} \,\mathrm{d}V + \int_{\partial V} \rho \underline{\mathbf{v}} \otimes \underline{\mathbf{v}} \cdot \mathrm{d}\underline{\mathbf{a}} = &- \int_V \nabla p \,\mathrm{d}V + \int_V \rho \underline{\mathbf{g}} \,\mathrm{d}V + \int_{\partial V} \left[\left(\underline{\underline{\mathbf{T}}} + \underline{\underline{\mathbf{T}}}^T \right) \right] \cdot \mathrm{d}\underline{\mathbf{a}} + \int_V \underline{\mathbf{f}}_b \,\mathrm{d}V \\ &- \sum_i \int_{\partial V} \alpha_i \rho_i \underline{\mathbf{v}}_{d,i} \otimes \underline{\mathbf{v}}_{d,i} \cdot \mathrm{d}\underline{\mathbf{a}} + \int_V \underline{\mathbf{S}}_i^\alpha \,\mathrm{d}V \end{aligned} \tag{1.11}$$

where

- p is the pressure,
- $\underline{\mathbf{g}}$ is the gravitational force vector,
- $\underline{\underline{\mathbf{T}}}$ is the viscous stress tensor,
- $\underline{\mathbf{f}}_b$ is the body force vector,

- $\underline{\mathbf{S}}_i^{\alpha}$ is the user defined phase momentum source term.

The **energy conservation** equation solved is

$$\frac{\partial}{\partial t}\int_V \rho \mathrm{E}\,\mathrm{d}V + \int_{\partial V}\left[\rho \mathrm{H}\underline{\mathbf{v}} + \sum_i \alpha_i \rho_i \mathrm{H}_i \underline{\mathbf{v}}_{d,i}\right]\cdot \mathrm{d}\underline{\mathbf{a}} = -\int_{\partial V}\underline{\dot{\mathbf{q}}''}\cdot \mathrm{d}\underline{\mathbf{a}} + \int_{\partial V}\underline{\underline{\mathbf{T}}}\cdot\underline{\mathbf{v}}\,\mathrm{d}\underline{\mathbf{a}} + \int_V \underline{\mathbf{f}}_b\cdot\underline{\mathbf{v}}\,\mathrm{d}V + \int_V \mathrm{S_E}\,\mathrm{d}V \tag{1.12}$$

where

- E is the total energy,
- H is the total enthalpy,
- $\underline{\dot{\mathbf{q}}''}$ is the heat flux vector,
- $\underline{\underline{\mathbf{T}}}$ is the viscous stress tensor,
- $\mathrm{S_E}$ is the user defined energy source term.

1.2 Modelling Turbulence

Flows are generally turbulent. Turbulence is chracterised by irregular fluid motion dominated by eddies and vortices across a wide range of spatial and temporal scales. The largest eddies extract kinetic energy from the mean flow, which is then, through a cascade, transferred to the eddies at the smallest scale where the energy is dissipated due to viscosity, raising the internal energy of the fluid [1, 2].

RANS models result from decomposing flow variables into mean and fluctuating components, $\phi = \overline{\phi} + \phi'$, applying Reynolds averaging, which filters out turbulent fluctuations below the averaging scale [3, 4]. This operation yields additional Reynolds stress terms representing momentum transport by turbulence, which require closure. In order to draw closure relations, eddy-viscosity models of turbulence assume that:

- the turbulent scalar flux vector aligns with the mean scalar gradient.
- the Reynolds stress tensor is proportional to the mean velocity strain rate through an eddy viscosity.

While these assumptions simplify modelling, they tend to be, more often than not, invalid as they do not fully capture turbulence physics, especiialy in anisotropic or non-equilibrium flow [5]. This limitation can be significant in regions with complex strain rates or strong secondary flows.

Although turbulence can be modelled in various ways to various levels of resolutions, turbulence in this study has been modelled using the Shear Stress Transport (SST) k-w model. The SST k-w model is a two-equation eddy-viscosity turbulence model based on the Reynolds-Averaged Navier-Stokes (RANS) equations. The model combines the advantages of the k-ω model near walls—which accurately resolves boundary layer behaviour—and the k-ε model in the free stream, enhancing robustness and accuracy in separated flows [6]. The SST approach uses a blending function to smoothly transition between these models, which is especially effective in flows with strong adverse pressure gradients and separation, such as multiphase cavity flows [7]. Another blending function limits the eddy viscosity in wake regions and a cross-diffusion term ensures consistency between the k-ω and k-ϵ models.

Since the form of the RANS governing equations remains similar to the governing equations described in the previous section, they are not explicity written here but can be found in the Star-CCM+ documentation [star-ccm]. The additional equations accounting for the transport of the turbulent kinetic energy and dissipation rate is given in 1.13

$$\frac{\partial \rho k}{\partial t} + \nabla(\rho k \underline{\overline{\mathbf{v}}}) = \nabla\left[(\mu + \sigma_k \mu_t)\nabla k\right] + P_k - \rho\beta^* f_{\beta^*}(\omega k - \omega_0 k_0) + S_k, \tag{1.13a}$$

$$\frac{\partial \rho \omega}{+}\nabla(\rho \omega \underline{\overline{\mathbf{v}}}) = \nabla\left[(\mu + \sigma_\omega \mu_t)\nabla \omega\right] + P_\omega - \rho\beta f_\beta(\omega^2 - \omega_0^2) + S_\omega, \tag{1.13b}$$

The turbulent viscosity, μ_t, is calculated as $\mu_t = \rho k T t$, where T_t is the turbulent time scale. where σ_k and σ_ω are the Prandtl numbers for k and ω respectively.

The implementation of th model in Star-CCM+ can be further controlled to adjust the realisability. Realisability is the postive semi-definiteness of the reynolds stress tensor. The specification of turbulence is done using the turbulence intensity and the length scale. The inherent anisotropy in turbulence is accounted for with quadratic and cubic constitutive options in the modelling of reynolds stresses. The parameters in the model are left on their default values as they are empirically modelled.

1.3 Evaporation

The evaporation process is distinguished between occuring at saturation temperatures or not. Evaporation at saturation temperatures, known as vaporisation, is a much faster process in comparison to evaporation at normal [8]. In case of vaporisation along a heated wall, most CFD models of vaporisation treat the energy transfer by partitioning the heat flux at the wall into the latent heat of the bubble vaporisation process and the sensible heat transferred to the liquid through conduction, convection, and quencing processes. This is determined by the area fractions covered by the liquid and vapour. Correlations between the heat flux and wall superheat [what is superheat] have been made using empirical results or based on modelling of sub-processes.

The sub-processes that are primarily taken into account in such determinations are:

- number density of nucleation sites
- diameter of bubble at time of detachment from the surface
- frequency of bubble detachment

Nucleation affects the number and distribution of bubbles in the fluid. Walls, particularly so in case they're heated, are more suspectible to be nucleation sites for bubble growth as the activation energy for bubble growth is much lower. This nucleation can be modelled through the domain as homogenous or heterogenous - where nucleation sites are modelled using a presumed PDF function or be determined by empirical correlations. [9]

After a stable bubble nucleus has been formed, the growth of the bubble becomes pertinent in the evaluation of the diameter at departure and the interphase energy and mass transfer. In the case of an ideal spherical bubble undergoing expansion in a uniformly superheated liquid, the rate of growth is determined by the surface tension, liquid inertia, and the differnce in pressures inside and outside the bubble. Surface tension dominates and restricts the initial bubble growth. This initial rapid expansion of the bubble results in the formation of a thin liquid layer separating the vapour from the wall surface. This microlayer can be a significant contributor to the bubble growth and heat transfer process. As the radius increases and the liquid layer around the bubble cools due to evaporation, a decrease in vapour pressure in seen. The dynamics are controlled by thermal diffusion and inertia. Inertial effects, however, become less relevant as the bubble growth rate decreases and the bubble radius grows. The interphase energy exchange determines further bubble growth. [10]

1.3.1 Effects of low pressure and gravity conditions

The absence of significant gravitaional forces alters the dynamics of vaporisation - affecting natural convection, buoyancy driven phenomenon, and bubble dynamics. Under terrestrial gravity, convection currents develop under thermal gradients, thus bringing cooler liquid to the heated surface as the vapour bubble rises due to buyoancy. This density driven natural convection is affected in low gravity conditions, maintaining the thermal gradient. Bulk fluid movement is affected and heat and mass transfer processes shift from being convection dominated to being diffusion dominated. [11, 12]

Under low gravity conditions, vapour bubbles tend to remain near the heat source, growing signifcantly larger and coalescing with smaller bubbles. Prolonged bubble presence on the surface results in drying out of the surface and unsteady rise in surface temperatures.[13] [Mention some numbers] In conditions where the dominance

of surface tension forces becomes more pronounced, the surfae tension gradients can induce fluid circulation. This is known as Marangoni convection and is observed to affect bubble growth and heat transfer. [14]

Reduced system pressure impacts the overall intensity of heat transfer. The onset of nucleate boiling also requires higher superheat values in low pressure conditions. Decrease in pressure leads to a lower number of active nucleation sites and increases microlayer thickness found at the bubble-wall interface. The local heat flux density from microlayer evaporation is impacted and the microlayer evaporation area and depletion time is increased. The system is also susceptible to the formation of a continuous vapour film at the surface due to coalescing bubbles unable to escape the surface, reducing the overall heat transfer. [15]

The affected heat transfer and bubble growth phenomenon further result in lower critical heat fluxes and heat transfer efficiency. [16, 17]

1.3.2 Vaporisation in a superheated liquid

[————UNDER CONSTRUCTION————————–]
Flash evaporation is the intense vaporisation triggered through depressurisation of superheated fluids.

1.4 Flow Regimes and Closures

Additional constitutive relations are required to 'close' the set of equations being solved that compensate for the various scales blended out due to the averaging procedures. Along with the equations of state that govern the physical properties of the materials involved, closure relations are required to account for the various interfacial momentum or energy transfer mechanisms which are dependent on the interfacial topology. The distribution of the phases cannot be accurately estimated a priori [18] but the macroscopic behaviour of the system is dependent on these small-scale interactions of the phases [19]. The existence of deformable interfaces leads to more complications.

The various flow regimes in two-phase gas-liquid flows cover the dispersed regimes to the presence of free surfaces. Some simplification is possible by grouping the various interfacial distributions together. Through the perspective of vertical tubes, this ranges from bubbly flows - where gas small gas bubbles are dispersed through the liquid- to annular flows - where the gas in the middle of the cross section and the liquid film at the walls are separated by a free surface. [18] Star-CCM+ in its implementation of the eulerian-multiphase framework utilises a similar dynamic flow regime topology based on the local volume fractions of the interacting phases. Based on the increasing volume fraction of the secondary phase, α_s, in a finite volume cell, the topology transitions through three principal regimes [20] [starccm], where the secondary phase is:

- dispersed in the primary phase (first dispersed regime, $0 \leq \alpha_s \leq \alpha_{fr}$)
- separated from the primary phase through a large scale interface (interfacial flow regime, $\alpha_{fr} \leq \alpha_s \leq \alpha_{sr}$)
- continuous while having the primary phase dispersed (second dispersed regime, $\alpha_{sr} \leq \alpha_s \leq 1$)

The first and second regime terminuses, α_{fr} and α_{sr}, mark the values of α_s that defines the end of the first dispersed and large scale interface regimes respectively. In the case of EMP, Star-CCM+ uses a regime-specific closure model for the interfacial momentum and energy terms. These models are blended using a weighting scheme according to the calculated proportion of each regime in the cell. The transport of the volume fraction is done through HRIC, ADIS, or normal 1st or 2nd order convection.

"neither of the two methodology - VOF or two-fluid - is suitable for multiscale flows [20]". "most data for the semi-empirical relations covers rather narrow domains [18]" "if one considers a non flow situation in the two horizontal layers, the forces acting on the two fluids and their interface are gravity and surface tension - in this case we speaf of the RT instability [21]." "there exists a wide spectrum of scales, ranging from the global one dictated by the flow configuration, to the smalles one where turbulent energy is dissipated into heat[22]"

1.5 Closures

Closures are critical to the accurate prediction of the relevant flow variables and the distribution of volume fraction [23]. Existing literature reviews the overwhelming number of closures available that yield a range of varyinh performance along with their comparison with experimental data, however literature presenting CFD simulations tend to choose 'result-oriented' closures that aren't transferable to a wider range of flow situations [24, 25]. Literature admits that the industry relies on the tunable parameters in the closures to match the specific experimental data being simulated or the arbitrariness of this choice to get improved flow predictions [[20] rzehak, kriebitzsch; silva [21] (sugrue)].

Models may also provide satisfying predictions for certain parameters or situations while failing in others and a general inconsenus prevails while discussing the effects of interfacial forces. Early work has tended to neglect the effects of lift or virtual mass forces, or chosen to use constant coefficients to describe them while using formulations for the drag forces based on relevant non-dimensional numbers. [26] The effect of non-drag forces, however, tends to be as significant on the distribution of phases. [liao2014-flashboiling] Since it is difficult to separate the effects of the various forces in actual experiments, the development of these closures assumes certain idealisations when they're not partly or wholly based on empirical data, rendering them highly flow dependent [troshko and hassan, [15] in Sugrue]. The authors of [troshko and hassan] critiques the methodology of resolving a complex interfacial momentum transfer through a linear superposition of several simple forces as the basis of the discrepancy.

The industry has, however, attempted to establish generality that can be transferred to multiple flow situations and has proposed baseline frameworks. There is a general consensus that the role of turbulence is underestimated [27]; current models also aim to reduce uncertainty and inconsistency in model selection [27], adopt closures that avoid "case-specific adjustments to unphysical coefficients and tunable parameters" [23], and utilise more physics based closures that are more physically consistent [26, 27].

Drag force is generally more dominant than other forces as they tend to be a magnitude higher than other forces [28]. Most drag models perform equally well at low gas volume fractions. However as they are based on ideal conditions, they do not account for bubble swarm effects at higher gas volume fractions and show reduced accuracy. Certain swarm corrections do exist, but they are averaged along a radial profile when the swarm effects are also felt in the bulk. [23, 25, 27] [is there any other update to corrections?] The accurate modelling of interaction area density and length becomes relevant for the calculation of the drag force. Using the linearised drag coefficient, A_D, and the relative velocity between the phases $\underline{\mathbf{v}}_r = \underline{\mathbf{v}}_j - \underline{\mathbf{v}}_i$, Star-CCM+ implements the drag force as,

$$\underline{\mathbf{F}}_{ij}^D = A_D \underline{\mathbf{v}}_r \tag{1.14}$$

The linearised drag coefficient is related to the standard definitions of the drag coefficient C_D using

$$A_D = C_D \frac{1}{2} \rho_c |\underline{\mathbf{v}}_r| \frac{a_{cd}}{4} \tag{1.15}$$

where a_{cd} is the interfacial area density and $a_{cd}/4$ represents the projected area of the equivalent spherical particle.

The lift force models are particularly sensitive and uncertain in their performance [25]. They suffer from the same discrepancy between performance in low and high gas volume fractions. Moreover, existing lift force implementations in Star-CCM+ fail to replicate the experimental data they were derived from. [23] Assuming α_p is the volume fraction of the primary (continuous) phase, Star-CCM+ implements the lift force as

$$F_{ij}^L = C_{L,\text{eff}} \alpha_d \rho_p \left[\underline{\mathbf{v}}_r \times \left(\nabla \times \underline{\mathbf{v}}_p \right) \right]. \tag{1.16}$$

A turbulent dispersion force is supposed to act on the gas phase and spreads this dispersed phase due to turbulence in the continuous phase. This is typically modelled as as a diffusion term proportional to the turbulent kinetic energy gradients [29]. The force is theorised to arise from the averaging of the fluctuations of the bubbles due to the turbulence.

$$F_{ij}^{TD} = A_D \underline{\mathbf{v}}_{TD} = A_D \underbrace{C_0 \frac{\nu_c^t}{\sigma_\alpha} \underline{\underline{\mathbf{I}}}}_{\underline{\underline{\mathbf{D}}}_{TD}} \cdot [\nabla \ln(\alpha_d) - \nabla \ln(\alpha_c)] \tag{1.17}$$

The closure relation by Burns [burns] uses the Favre averaging of the drag force and can be expressed as

$$F_{ij}^{TD} = C_D C_{TD} \frac{\mu_l^t}{\rho_c \sigma_{TD}} \underline{\mathbf{v}}_{TD}, \tag{1.18}$$

where C_D is the drag coefficient, C_{TD} is an adjustable turbulent dispersion calibration coefficient, μ_l^t is the liquid turbulent viscosity, and σ_{TD} is the turbulent Prandtl number.

Most lift coefficient correlations don't account for the decrease of the magnitude of the lift force to zero near the walls. This low gas volume fraction at the walls has historically been explained through a liquid film forming at the walls repelling the bubbles. This has been demonstrated to not actually occur in practice. The wall lubrication force has thus been criticised as lacking any physical justification. [23, 27] Newer closures [lubchenko] utilise a turbulent dispersion force based formualtion to maintain the desired gas volume fraction and use the closure instead as a correction term to resolve the phase distribution at the boundary lost due to the averaging process. Star-CCM+'s implementation of the wall lubrication force

$$F_{ij}^{WL} = -C_{WL}(y_w)\alpha_s \rho_p \frac{|\underline{\mathbf{v}}_r - (\underline{\mathbf{v}} \cdot \underline{\mathbf{n}})\underline{\mathbf{n}}|}{d_b} \underline{\mathbf{n}} \tag{1.19}$$

can be changed to implement alternative models instead by defining user defined field functions for C_{WL}.

Surface tension, although an interfacial force, is implemented using the Continuum Surface Force approach as a volumetric force in the VOF formulation. Divided into normal and tangential components and using the mean curvature of the free surface $\kappa = -\nabla \cdot (\nabla \alpha_i / |\nabla \alpha_i|)$, the volumetric surface tension force is given as

$$\underline{\mathbf{f}}_\sigma = \underbrace{-\sigma \nabla \cdot \frac{\nabla \alpha_i}{|\nabla \alpha_i|} \nabla \alpha_i}_{\underline{\mathbf{f}}_{\sigma,n}} + \underbrace{(\nabla \sigma)_t |\nabla \alpha_i|}_{\underline{\mathbf{f}}_{\sigma,t}} \tag{1.20}$$

The EMP treats surface tension by splitting it among the phases occupying the cell using the local volume fractions. The pressure gradient is then the summation of the individual phase surface tension contributions $\underline{\mathbf{f}}_{\sigma,i}$, [20] [starccm],

$$\underline{\mathbf{f}}_{\sigma,i} = \alpha_i \sigma \kappa \nabla \alpha_i. \tag{1.21}$$

1.6 Closures: Under Construction

- Interphase mass transfer (evaporation/condensation) Represents phase change, modeled as interfacial flux determined by local thermodynamic properties (temperature, concentration, pressure—often using kinetic models for flash evaporation).
- Interphase energy transfer Encompasses latent heat and sensible heat exchange at the interface and throughout the phases (essential for accurate phase change modeling).
- Nucleation (wall and bulk)
- Interfacial area changes due to bubble growth, coalescence, breakup
- Local pressure and temperature gradients driving simultaneous mass and energy transfer

1.7 Numerical Methods

[————SOMEWHAT UNDER CONSTRUCTION————————]

Star-CCM+ is a commerical CFD software currently maintained and developed by Siemens Digital Industries

Software. The software uses the Finite Volume Method to discretise the integral form of the governing differential equations over a number of control volumes in the domain. The software has an automatic meshing system but allows the implementation of custom meshing using surface wrappers. The relevant variables are calculated at the center of each control volume and interpolated to the control volume surfaces. Convective and diffusive fluxes are interpolated using the second-order upwind differentiation scheme which uses the linear interpolation between two upwind centroids to approximate the fluxes at the surfaces of the control volume. Gradients are interpolated using the least squares method with Venkatakrishnan gradient limiters. While the VOF formulation uses HRIC, the EMP framework convects volume using either the first- or second-order upwind differentiation. In case the Large Scale Interface Detection model is active, HRIC or a scheme that blends HRIC with the first- or second-order upwind scheme can be implemented can also be chosen for volume fraction convection. The semi-implicit method for pressure-linked equations (SIMPLE) algorithm is used as the default segregated pressure-velocity coupling scheme.

1.8 Workflow?

[————UNDER CONSTRUCTION————————]

2 Subproblem: Liquid Infiltration

This chapter covers the simulation setup of liquid infiltration of the evaporation chamber. The physics models used in the simulation of the problem are validated against the 1D Stefan problem and the 2D Rayleigh-Taylor instability. The construction and result of these simulations is shortly discussed. The chapter concludes with the description of the computational methodology, results, and comments on the 2D simulation of the infiltration problem.

2.1 Problem Description and Objective

The first subproblem addresses water infiltrating the evaporation chamber during rocket ascent.

The flow system assembly comprises the hydraulic accumulator, the evaporation chamber, and the injection system subassembly that drives the liquid from the hydraulic accumulator into the evaporation chamber. The hydraulic accumulator and the evaporation chamber can be heated until lift-off. Current design considerations assume the temperature of the evaporation chamber and water in the accumulator to be 368.15 K and 353.15 K respectively. The ambient temperature at lift-off is assumed to be 280 K based on the remarks and considerations made in [OLD-SED].

Although insulation from the environment outside the rocket is assumed and the hydraulic accumulator will be sufficiently insulated, the evaporation chamber may cool down slightly during ascent until the start of the experiment at time $t = 140$ s after lift-off. As of yet, no considerations have been made regarding the insulation of the evaporation chamber.

The pre-charge pressure in the accumulator and the initial pressure in the evaporation chamber are assumed to be 3.5 bar and 1 bar respectively. Water injection begins after the data storage signal is sent by the electronics module and stops shortly before the apogee is reached, when the absolute pressure in the evaporation chamber rises to 1.7 bar. Due to water remaining at the time of cutoff in the injection subassembly and also the accumulator, the total amount of water discharged to the chamber is assumed to be 150 mL.

No flow measurements are made in the injection assembly. Pressure and temperature are measured in the evaporation chamber along the nozzle at the midspan, the throat, and the exit.

The objective of the first task is to model the water injection into the evaporation chamber and extract quantities of interest to better estimate the initial conditions for the second task. The validity of the multiphase physics models provided by Star-CCM+ and the equations implemented therein is also evaluated against the one-dimensional Stefan problem and the two-dimensional Rayleigh-Taylor instability.

2.2 Validation Problems

2.2.1 Stefan Problem

Phase change processes are characterized by the movement of an interface separating two distinct phases, where one phase expands as another contracts due to thermal or non-thermal mechanisms [30]. The mathematical model governing such moving boundary problems is known as the Stefan system [30], and the analysis of such a system forms the basis of the Stefan problem.

In the classical Stefan problem, a fixed-temperature boundary propagates a melting or freezing front into the domain, with interface movement determined by the balance of latent and conducted heat. [31] With uniform initial temperature, constant boundary temperature and thermophysical properties in each phase, the semi-infinite 1D Stefan problem is the simplest, explicitly solvable problem of phase change. As the problem admits an analytical solution and reduces to the standard transient heat conduction problem when latent heat is neglected, it serves as a rigorous benchmark for validating phase-change modelling and the correct implementation of the energy equation. [31, 32, 33].

Assuming that the region in space, $\Omega \subset \mathbb{R}^2$, occupied by the material is the 2D semi-infinite rectangle $\Omega = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \in [0, \infty), y \in [0, \Delta h]\}$, with the solid phase and the liquid phase occupying the regions Ω_s and Ω_l respectively such that $\Omega = \Omega_s \cup \Omega_l$. At any given time, t, the two phases are separated by the interface, $X(t)$, that marks the change in phase of the material. The boundary of the domain Ω is denoted by $\partial\Omega$.

The problem space, $\mathcal{D} = \{[t_0, t_f] \times \Omega\}$, is the spatial and temporal domain occupied by the material. With the heat equation governing the temperature field in the liquid region, the Stefan condition at the interface, and the respective boundary conditions, the system of equations defining the one-dimensional Stefan problem can be written as,

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}, \text{ for } 0 < x < X(t), t_0 < t < t_f \tag{2.1a}$$

$$-k \frac{\partial T(X(t), t)}{\partial x}\Big|_{X^-(t)} = \rho L \frac{\partial X(t)}{\partial t}, \text{ for } t_0 < t < t_f \tag{2.1b}$$

$$T(x, 0) = T_m, \text{ on } \Omega \tag{2.1c}$$

$$T(0, t) = T_l, \text{ on } \partial\Omega(0, t) \text{ for } t_0 < t < t_f \tag{2.1d}$$

$$\nabla T \cdot \underline{\mathbf{n}} = 0, \text{ on } \partial\Omega - \partial\Omega(0, t) \tag{2.1e}$$

An analytical solution to the system of equations is given as

$$\hat{T}(x, t) = T_l - (T_l - T_m) \frac{\text{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right)}{\text{erf}\left(\frac{X(t)}{2\sqrt{\alpha t}}\right)}, \tag{2.2}$$

$$\hat{X}(t) = 2\lambda\sqrt{\alpha t}, \tag{2.3}$$

where λ is the unique root of the monotonic function

$$f(\lambda) := \frac{C_p(T_m - T_l)}{L} - \frac{\lambda e^{-\lambda^2}}{\sqrt{\pi}\text{erf}(\lambda)}$$

The error function $\text{erf}(x)$ is defined as $\text{erf}(x) := 2/\sqrt{\pi} \int_0^x e^{-y^2}\, dy$. The physical parameters and computational methodology are summarized in Table 1

The simulations assume a laminar but unsteady flow. The computational domain of the simulation is chosen to be $[0, L] \times [0, 4\Delta h]$, where Δh is the cell width of the discretised domain. The domain is discretised using a trimmed cell mesher [CONFIRM] with uniform quadrilateral elements. The volume of fluid method with first order HRIC is used to reconstruct the interface. The convection and diffusion terms are discretised using the second-order upwind scheme and the least squares method was used to interpolate the values of the gradients from the cell centres onto the face centres. The transient term is discretised using the implicit Euler-method which is first-order accurate. All four boundaries are assigned a no-slip shear stress specification. The boundary at $x = 0$ is assigned a constant temperature, $T_l = 300$ K, while the other boundaries are assumed to be adiabatic. Since Star-CCM+ sets under- and over-relaxation factors conservatively, the values are left unchanged.

The physical parameters and the computational methods involved in the simulation setup are summarised in Table 1. The evolution of the temperature field for the simulation with cell size, $\Delta h = 5.0 \times 10^{-3}$ m is shown in Figure 1. The interface is observed to move towards the right with increasing simulation time. The temperature of the system can also be seen to decrease from the Dirichlet boundary at $x = 0$ towards the melting temperature, T_m, at the interface, after which it stays constant.

Simulation Setup	
Domain	2D rectangular domain
Mesh size, Δh	2.0×10^{-3} m
Time discretization	Implicit unsteady, 1st order Euler
Time step, Δt	0.5 s
Physical Properties	
Density, ρ	$1000.0\,\mathrm{kg{\cdot}m^{-3}}$
Dynamic viscosity, μ	8.8871×10^{-4} Pa·s
Specific heat, C_p	$4200.0\,\mathrm{J{\cdot}(kg{\cdot}K)^{-1}}$
Latent heat of fusion, L	$1.0 \times 10^{5}\,\mathrm{J{\cdot}kg^{-1}}$
Thermal conductivity, k_l	$2.33\,\mathrm{W{\cdot}(m{\cdot}K)^{-1}}$
Thermal conductivity, k_s	$0.62\,\mathrm{W{\cdot}(m{\cdot}K)^{-1}}$
Melting temperature, T_m	273 K
Numerical Methods	
Multiphase model	VOF with HRIC convection
Flow model	Laminar
Spatial discretization	2nd order (energy), 1st order (HRIC)

Table 1: CFD simulation parameters for Stefan melting-solidification problem

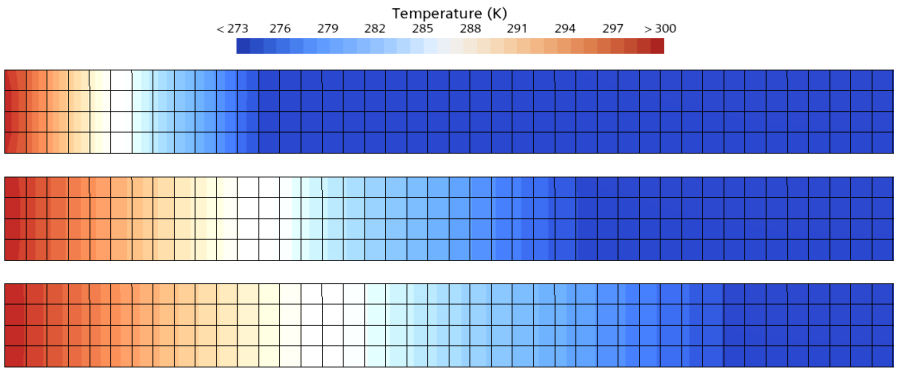


Figure 1: Evolution of the phase change interface

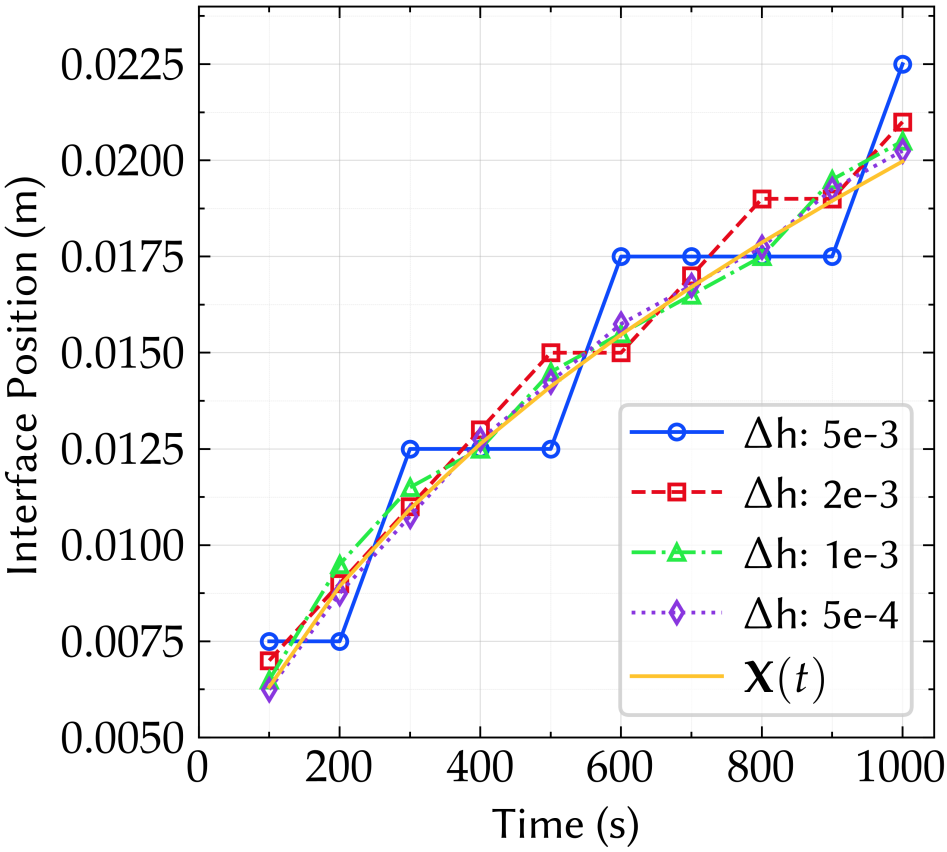
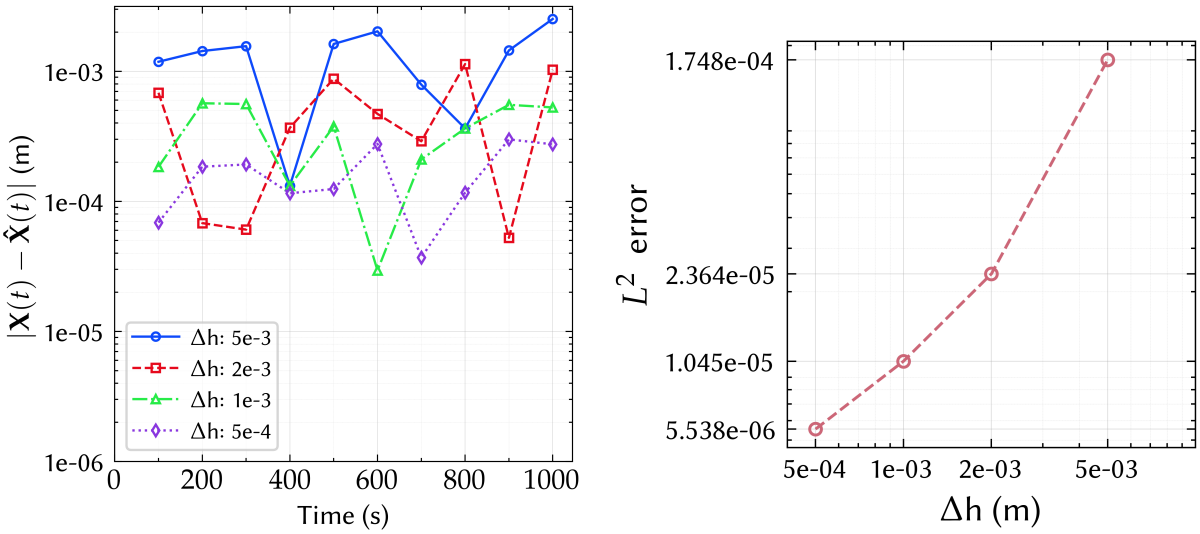


Figure 2: Location of the phase change interface



(a) Absolute error in the location of the interface

(b) Temperature profile convergence analysis

Figure 3: Error Analysis for the Stefan problem for different mesh sizes

Figure 3 shows the location of the interface for simulations with different cell sizes against the analytical solution. The location of the interface is inferred using the volume fraction of the liquid and solid phases present along with the temperature results. The numerical values are plotted at time intervals of 100 s and the absolute error in the location of the interface can be seen in Figure 3a. The temperature profiles obtained numerically can

also be compared against the analytical temperature profile in order to calculate the L_2 and L_∞ errors, where $L_2 = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(T_i - T_{analytical,i})^2}$ and $L_\infty = \max_i |T_i - T_{analytical,i}|$ and evaluate the convergence rate.

The errors in the temperature profiles for the various mesh sizes are plotted in Figure 3b. It can be seen that the simulation results converge to the analytical solution with decreasing mesh size.

2.2.2 Rayleigh-Taylor Instability

The Rayleigh-Taylor (RT) instability occurs when a denser fluid is accelerated into a lighter one, leading to interfacial perturbations and complex vortex formation [34]. This phenomenon, known as the Rayleigh-Taylor (RT) instability, is fundamental to many natural and engineering systems. Although the ideal, undisturbed RT instability is rarely encountered in practice, its essential features arise across a broad range of spatial and temporal scales—where the basic mechanism can be influenced by viscosity, surface tension, compressibility, external accelerations, or stratification[35]

Perturbation growth is driven by oppositely directed gradients of pressure and density at the interface. The density contrast between the two fluids is represented by the non-dimensional *Atwood number*, $At = (\rho_H - \rho_L)/(\rho_H + \rho_L)$, where ρ_H and ρ_L denote the densities of the upper (heavy) and lower (light) fluids respectively. The Atwood number quantifies the strength of the instability, with larger values corresponding to greater density contrasts.

The nonlinear development of the RT instability is central to a wide class of interfacial flows where one fluid penetrates or displaces another [36]. Accurately simulating RT instability serves as a classical benchmark for two-phase flow solvers, requiring detailed computation of interface dynamics and rigorous implementation of interphase interaction terms as discussed in [chapter stuf], and is widely used to validate numerical methods for multiphase flows [37].

Considering a region, $\Omega \in \mathbb{R}^2$, where two immiscible, viscous, and incompressible fluids occupy the regions Ω_H and Ω_L, such that $\Omega = \Omega_H \cup \Omega_L$. The regions $\Omega_{H,L}$ are defined using a scalar height function $h(x,t)$ that graphs the vertical position of the interface at each point along the x-axis.

$$\Omega_i(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : (-1)^i(y - h(x,t)) > 0, t \geq 0\},\ i = H, L$$

The interface, $X(t)$, separating the two immiscible fluids is given as

$$X(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : y = h(x,t),\ t \geq 0\}.$$

With the initial position of the interface denoted by $X(t = t_0) = X_0$, and the initial velocity field by $\underline{\mathbf{v}}(t = t_0) = \underline{\mathbf{v}}_0$, the flow is governed by the incompressible Navier-Stokes equations, surface tension σ, and acceleration $\underline{\mathbf{g}}$. The jump conditions at the interface preserve the velocity field and relate the normal stresses at the interface to the mean curvature.

If density ρ can be assumed to be $\rho_{H,L}$ or the mixture density in cases of the two-fluid and VOF methods respectively. The same can be extended to the velocity, $\underline{\mathbf{v}}$ and the dynamic viscosity μ.

The system of equations governing this two-phase flow system can be expressed as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \underline{\mathbf{v}} = 0 \qquad \text{in } \Omega_i(t) \tag{2.4}$$

$$\frac{\partial (\rho \underline{\mathbf{v}})}{\partial t} + \nabla(\rho \underline{\mathbf{v}} \otimes \underline{\mathbf{v}}) = -\nabla p + \mu \nabla^2 \underline{\mathbf{v}} + \rho \underline{\mathbf{g}} \quad \text{in } \Omega_i(t) \tag{2.5}$$

$$\nabla \cdot \underline{\mathbf{v}} = 0 \tag{2.6}$$

$$[[\underline{\mathbf{v}}]] = 0 \tag{2.7}$$

$$[[\left(\mu(\nabla\underline{\mathbf{v}} + (\nabla\underline{\mathbf{v}})^T) - (p + \rho\underline{\mathbf{g}})\underline{\underline{\mathbf{I}}}\right)\underline{\mathbf{n}}]] = \sigma\kappa\underline{\mathbf{n}} + [[\rho]]\underline{\mathbf{g}} \tag{2.8}$$

$$\underline{\mathbf{v}}(\underline{\mathbf{x}}, t_0) = \underline{\mathbf{v}}_0 \tag{2.9}$$

$$X(t_0) = X_0 \tag{2.10}$$

where $\underline{\mathbf{n}}$ is the normal vector to the interface, p is the pressure field, $\underline{\underline{\mathbf{I}}}$ is the second-order identity tensor, and κ in this case denotes the mean curvature of the interface.

In order to simulate a single mode perturbance to the interface, the initial position of the interface is defined using the amplitude, $A = 0.1L_x$, and wavelength, $\lambda = L_x$ as follows

$$X_0 = 2L_x + A\cos\left(\frac{2\pi x}{L_x}\right)$$

The dimension L_x is the characteristic length which is chosen to be the width of the simulated domain.

The numerical simulations assume turbulent and unsteady flow. The computational domain is a two-dimensional rectangle, $[0, L_x] \times [0, 4L_x]$ that is discretised using uniform orthogonal quadrilateral cells. Convective and diffusive terms are discretized by the second-order upwind scheme; cell-center gradients are interpolated to face centers using the least-squares method. Interface capturing is performed with the volume-of-fluid (VOF) method using the first-order HRIC scheme. The transient term is approximated using the first-order implicit Euler method. Turbulence effects are modeled using the Shear Stress Transport (SST) k-ω [better word?] closure, and the surface tension force is treated semi-implicitly. Gravitational acceleration acts downward, $g = 1.0\,\mathrm{m/s^2}$. Periodic boundary conditions are imposed in the horizontal direction and slip walls at the top and bottom.

The DNS study undertaken in [38] is taken to be the canonical benchmark and the physical parameters are chosen to maintain dynamic similarity.

According to [39], a reasonable solution to the RT instability with $At \approx 0.75$ can be obtained with a 128×512 mesh. With similar mesh resolutions being used in [38, 40] to study similar RT instabilities, the domain is discretised using a uniform orthogonal quadrilateral mesh with size $\Delta h = 4.5 \times 10^{-3}$ m. Additional mesh sizes are evaluated in order to study mesh convergence. Time is non-dimensionalised using the Atwood number and the timestep for the simulations, similar to [41, 38], is selected as $\Delta t = 2.5 \times 10^{-4}/\sqrt{At}$ s.

The physical parameters and the simulation setup, along with the various cases studied are summarised in Tables 2 and 3.

Figure 4 depicts the evolution of the RT instability for Reynolds numbers 300, 1000, and 3000. The initial evolution of the interface is through exponential growth of the infinitesimal disturbance of the interface [42]. The growth follows linear stability theory which transitions into increasing non-linear behaviour as the amplitude grows to orders of 0.1λ [35]. This non-linear evolution is marked by the development of the distinct "bubbles" of lighter fluid rising as the "spikes" of the heavier fluid fall down. The evolving interface experiences increasing vorticity and the increasing velocity shear results in the development of Kelvin-Helmholtz type instabilities [38, 36] leading to the roll-up of the spikes as the amplitude grows to the order of $0.5 - 1\lambda$. For the case of $Re = 3000$, these instabilities are noticeable much earlier all along the interface, leading to a secondary perturbation along the boundary of the domain. Further growth leads to secondary vortices along the spike ends which leads to chaotic mixing and complex flow features. [35].

- **Effect of Reynolds and Atwood Number:** Increasing the Reynolds number has a relatively small effect on the penetration lengths of the spikes and bubbles - with $Re = 300$ showing the least penetration depths. However, it does have a significant effect on the topology of the interface, with increasingly chaotic vortices visible at the spike rollup and the ensuing turbulent mixing. At $Re = 300$, it is seen in Figure 4, that the flow remains relatively laminar with smooth interfaces, whereas $Re = 3000$ shows significant turbulent mixing and vortical complex structures. High Atwood numbers promote non-linearity and increased turbulent mixing, and affect the initial interface growth. The effect and prominence of the counter-rotating vortices at the tip of the spike decreases and reduces the chaotic flow features at the tips

Spatial and Temporal Discretisation	
Domain	2D, $[0, L_x] \times [0, 4L_x]$, where $L_x = 1\,\mathrm{m}$
Time discretization	Implicit unsteady, 1st order
Time step, Δt	$3.5 \times 10^{-3}\,\mathrm{s}$
Physical Properties	
Amplitude of perturbation, A	0.1
Wavelength of perturbation, λ	$L_x = 1.0\,\mathrm{m}$
Surface tension, σ	$1.0 \times 10^{-5}\,\mathrm{N{\cdot}m^{-1}}$
Gravity, $\underline{\mathbf{g}}$	$[0, -1.0]\,\mathrm{m{\cdot}s^{-2}}$
Numerical Methods	
Multiphase model	VOF with HRIC convection
Flow model	Turbulent, SST k–ω
Spatial discretization	2nd order (flow), 1st order (HRIC)
Gradient reconstruction	Least-squares

Table 2: CFD simulation parameters for Rayleigh-Taylor instability study

	At = 0.2			At = 0.5			High At		
Parameter	Re300	Re1000	Re3000	Re300	Re1000	Re3000	At30	At1000	At1500
ρ_H [kg·m^{-3}]	1.5	1.5	1.5	3.0	3.0	3.0	3.0	1.0	1.5
ρ_L [kg·m^{-3}]	1.0	1.0	1.0	1.0	1.0	1.0	0.1	0.001	0.001
$\mu_{H,L}$ [Pa·s]	0.005	0.0015	0.0005	0.01	0.003	0.001	0.001	0.001	0.001
Mesh size Δh [m]	0.0045	0.0045	0.0045	(0.01, 0.007, 0.0045)	0.0045	(0.01, 0.007, 0.0045)	0.0045	0.0045	0.0045

Table 3: Fluid properties and mesh resolutions for all simulation cases

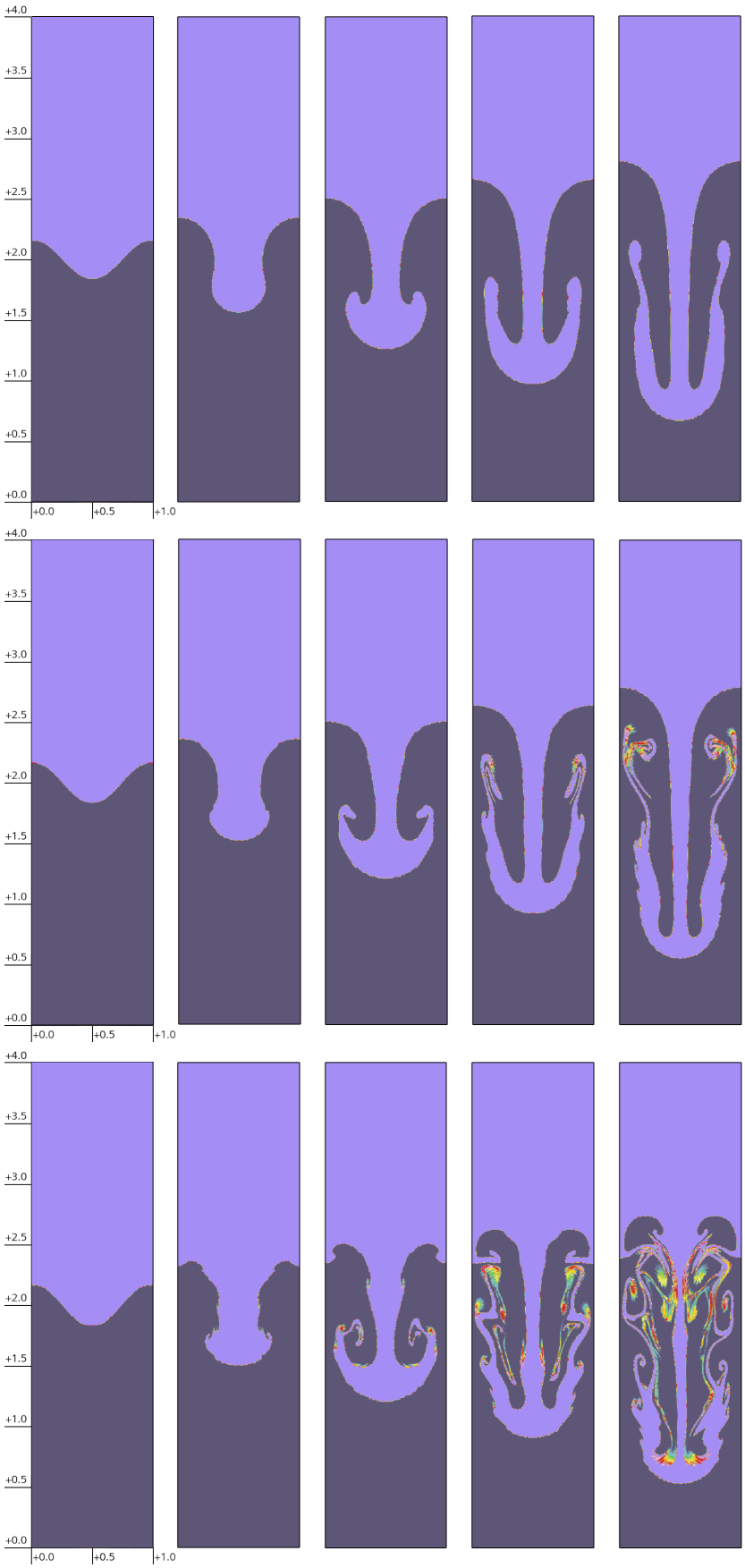


Figure 4: Evolution of Rayleigh-Taylor instability for Atwood number At = 0.5 at different Reynolds numbers.

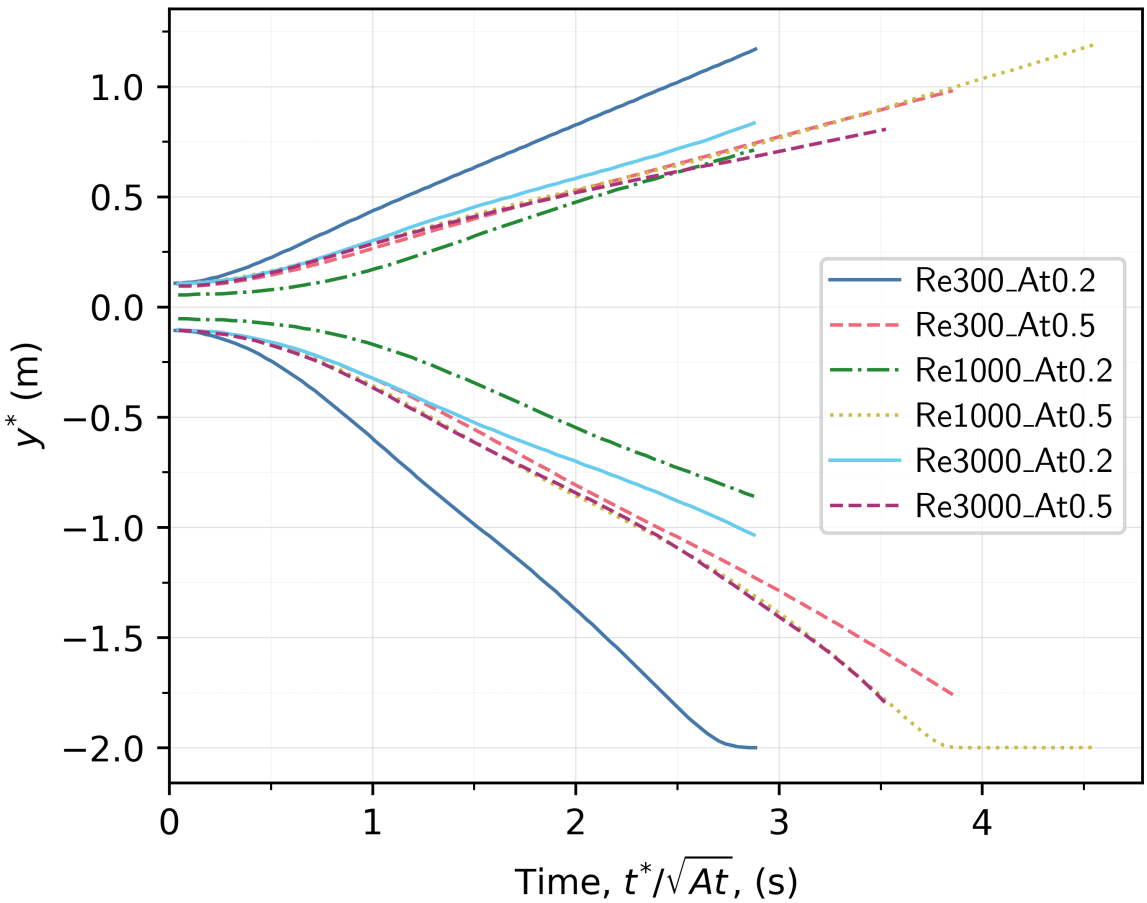


Figure 5: Evolution of the RT instability with varying Reynolds and Atwood numbers

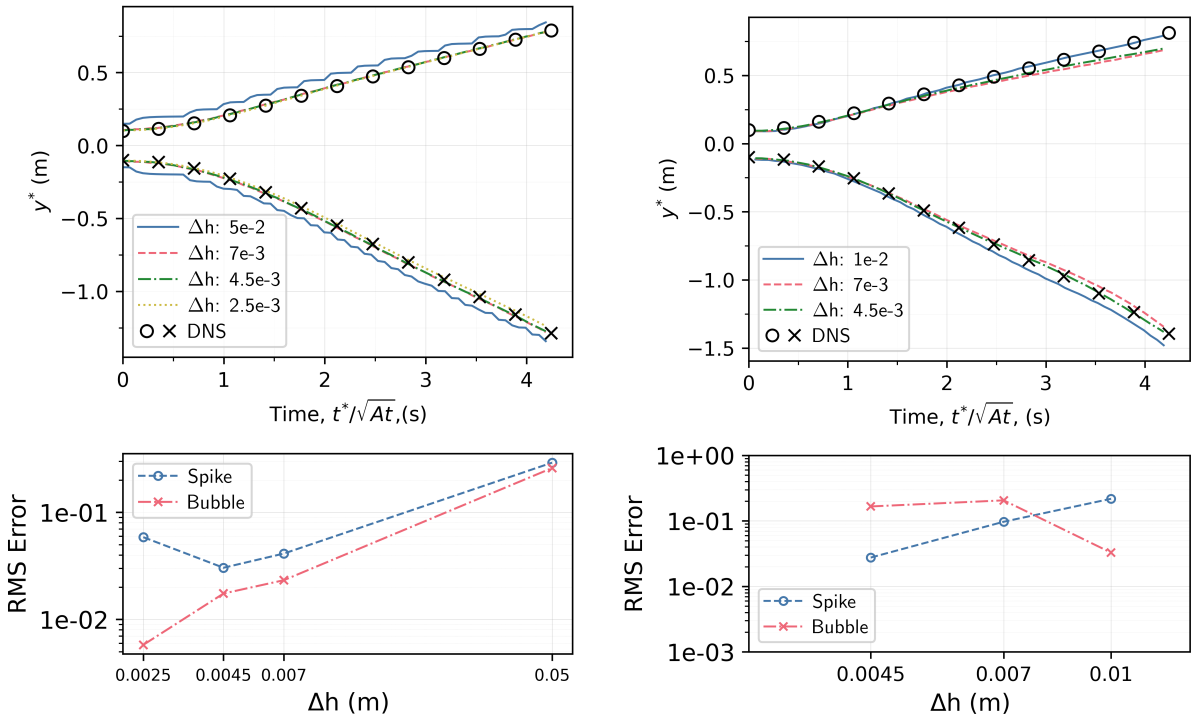
of the secondary vortices. The spike also falls much quicker. This is due to the density-driven acceleration outweighing the viscous effects as is seen in Figure 8 and in the very high density ratio simulations of the RT instability in [40].

- **Grid Convergence:** In order to evaluate the quality of the mesh, the maximum vertical distance of the spike and bubble tip as measured from the undisturbed initial interface, $y^* = |y_{s,b} - y_0|$, are evaluated for Reynolds numbers, $Re = 300, 3000$, and Atwood number, $At = 0.5$. Locations of the spike and bubble tips are recorded at every time-step and extrapolated in order to compare to the DNS data available from [38].

Figure 6 shows the comparison the results of the simulations with the DNS location of the bubble and spike tips along with the root mean square defined as $err = \sqrt{\sum_{t_0}^{t_f}(y_b - y_{dns})^2}$. The RMS errors and the maximum absolute errors are further summarized in table [table].

Figure 7 compares the RT instabilities for $Re = 1000, 3000$ and $At = 0.5, 0.2$ respectively without the evaluation of the errors.

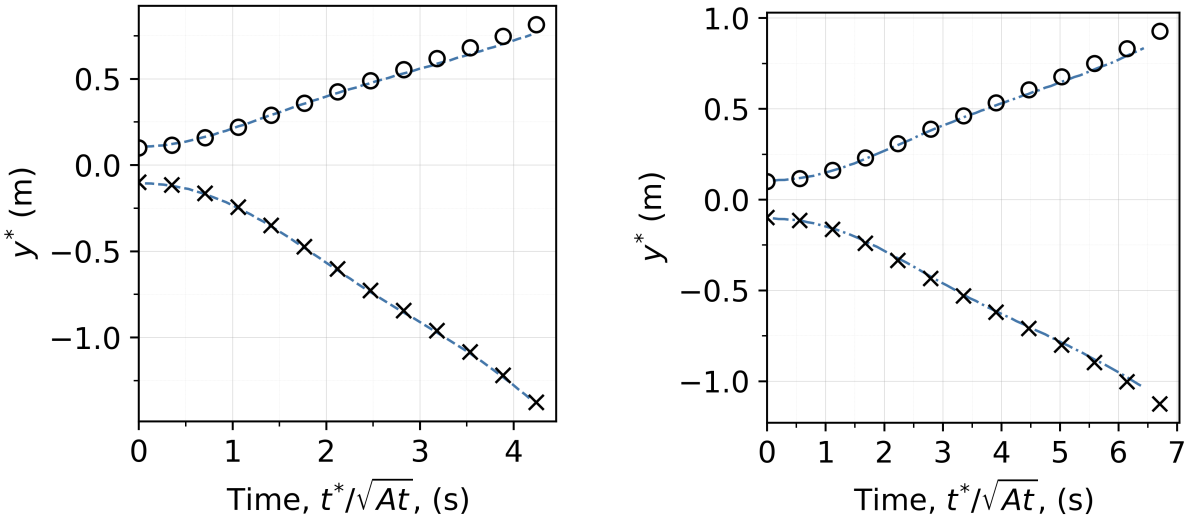
Although, Star-CCM+, qualitatively, accurately simulates the RT instability, the calculated RMS and absolute errors in the bubble and spike tip locations show erratic behaviour with decreasing mesh size. For $Re = 300$, the sudden increase in the RMS error at $\Delta h = 0.0025$ can be a result of [?] For higher Reynolds numbers, the bubble tip distances are consistently under-predicted for $Re = 1000, 3000$ at later simulation times. For $Re = 3000$, at later simulation times, secondary perturbations develop at the domain edges. These secondary perturbations develop into secondary spikes that interfere with the turbulent vortices at the spike tips and



(a) $Re = 300$

(b) $Re = 3000$

Figure 6: Grid convergence analysis for Rayleigh-Taylor instability



(a) $Re = 1000$, $At = 0.5$

(b) $Re = 3000$, $At = 0.2$

Figure 7: Comparison with DNS results for different Reynolds numbers and Atwood numbers

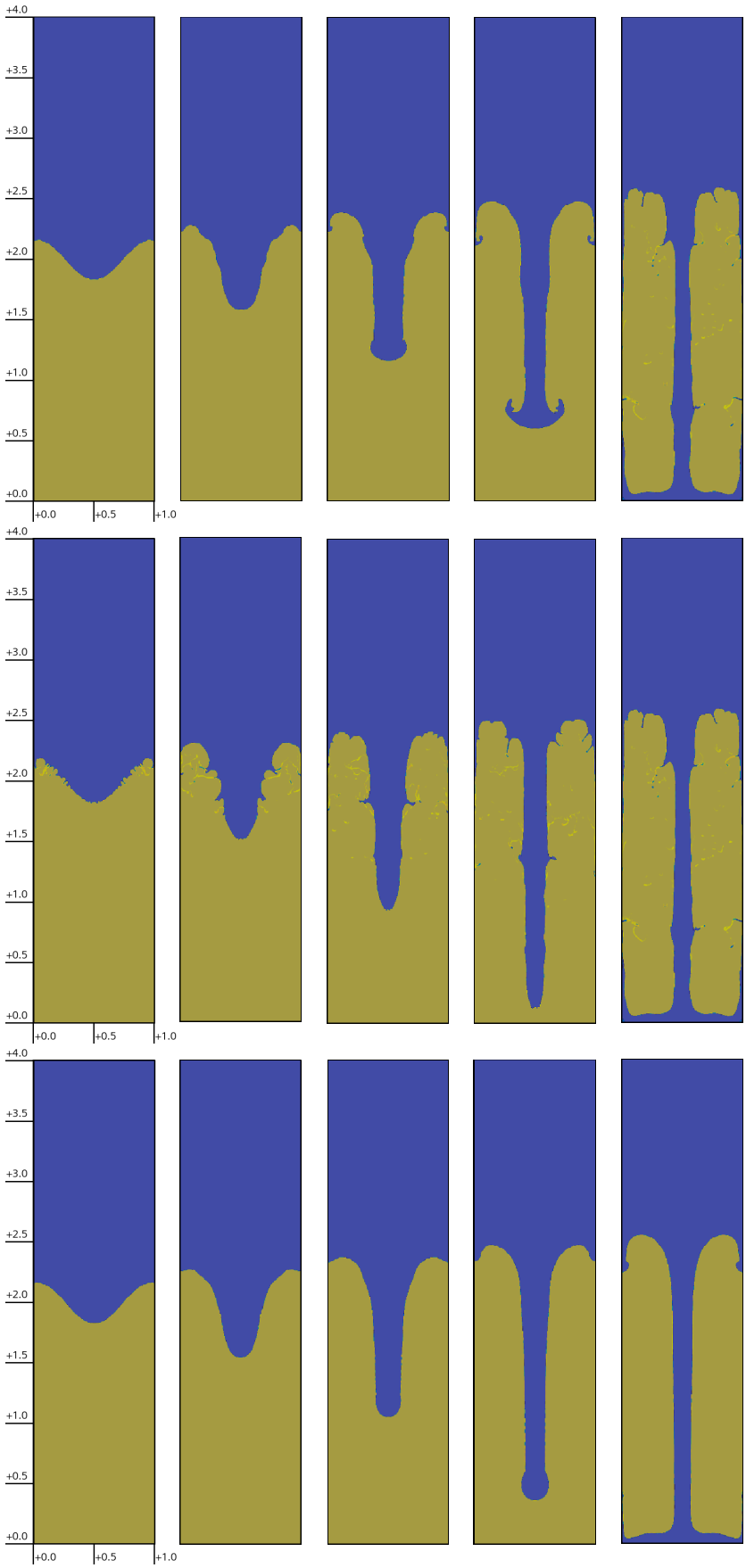


Figure 8: Evolution of Rayleigh-Taylor instability for high Atwood numbers At = 30, 1000, and 1500.

the main spike stem. This is, interestingly, not seen for the coarsest mesh size for $Re = 3000$. This could be developing Kelvin-Helmholz instabilities at the surface that transition into secondary RT instabilities [43]. Although Kelvin-Helmholz instabilities develop along the interface at higher Reynolds number, this secondary perturbation is first observed for $Re = 10000$ in [38].

This can be ascribed to either the lower order scheme used in the Star-CCM+ simulations or the choice of turbulence modelling using eddy viscosity models instead of completely resolving turbulence. For Atwood number that are not close to zero, the Boussinesq approximation is also invalid which is an integral assumption in RANS turbulence models. Star-CCM+, in order to accommodate the spurious currents that may be introduced through the use of eddy viscosity models, provides models that aid in the damping of momentum and/or turbulence at the interface, however, the addition of these models leads to significant deviations from the DNS results.

2.3 Liquid Infiltration of Evaporation Chamber

The evaporation chamber and the nozzle are modelled together as a 2D domain as shown in Figure 9. Line probes present along the mid-section of the evaporation chamber and at various points along the nozzle measure the flow quantities as would be measured in the actual experiment. Star-CCM+, during the creation of a 2D domain, prescribes an automatic, unchangeable depth of $1m$ to the model. It is assumed that approximately $150mL$ water is discharged into the evaporation chamber during the experiment. In order to maintain dynamic similarity, the mass flow rate is compared between the 3D experiment and the 2D model.

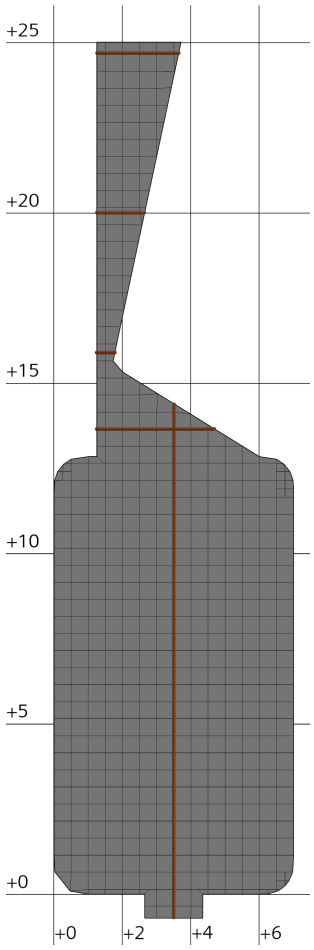


Figure 9: Computational mesh for the evaporation chamber and nozzle assembly

An adaptive trimmed cell mesh is used to mesh the computational domain.

- [ABOUT CONVERGENCE]
- [ON EVOLUTION OF INTERFACE and FLOW FIELD and RESULTING HEIGHT OF WATER]
- [ON EVOLUTION OF TEMPERATURE FIELD]

3 Subproblem: Evaporation

3.1 Objective

3.2 Validation Problems

3.2.1 Stefan Problem

3.3 Evaporation of Superheated Water in the Chamber

References

[1] Lindsey Anne Gilman. Development of a general purpose subgrid wall boiling model from improved physical understanding for use in computational fluid dynamics.

[2] Lars Davidson. An Introduction to Turbulence Models.

[3] D. C. Besnard and F. H. Harlow. Turbulence in multiphase flow. 14(6):679–699.

[4] Ch Hirsch. *Numerical Computation of Internal and External Flows: Fundamentals of Computational Fluid Dynamics*. Elsevier/Butterworth-Heinemann, 2nd ed edition, 2007.

[5] Stephen B. Pope. *Turbulent Flows*. Cambridge Univ. Press, 1. publ., 12. print edition.

[6] David C. Wilcox. *Turbulence Modeling for CFD*. DCW Industries, 3rd ed edition, 2006.

[7] F. R. Menter. Two-equation eddy-viscosity turbulence models for engineering applications. *AIAA Journal*, 32(8):1598–1605, 1994.

[8] Daniel Wickert and Günther Prokop. Simulation of water evaporation under natural conditions—A state-of-the-art overview. 3(4):242–249.

[9] Yixiang Liao and Dirk Lucas. Computational modelling of flash boiling flows: A literature survey. 111:246–265, 2017.

[10] Eran Sher, Tali Bar-Kohany, and Alexander Rashkovan. Flash-boiling atomization. 34(4):417–439.

[11] T H Cochran and J C Aydelott. Effects of fluid properties and gravity level on boiling in the discrete bubble region.

[12] Jeongmin Lee, Issam Mudawar, Mohammad M. Hasan, Henry K. Nahra, and Jeffrey R. Mackey. Experimental and computational investigation of flow boiling in microgravity. 183:122237.

[13] D. M. Qiu, V. K. Dhir, D. Chao, M. M. Hasan, E. Neumann, G. Yee, and A. Birchenough. Single-Bubble Dynamics During Pool Boiling Under Low Gravity Conditions. 16(3):336–345.

[14] Chun Long Xu, Yong Chao Shi, Xiao Xing Jin, Zuo Sheng Lei, Yun Bo Zhong, and Jia Hong Guo. Numerical Study of Marangoni Convection around a Single Vapor Bubble during Pool Boiling. 624:262–266.

[15] Vladimir Serdyukov, Ivan Malakhov, and Anton Surtaev. The Influence of Pressure on Local Heat Transfer Rate under the Vapor Bubbles during Pool Boiling. 16(9):3918.

[16] Hui Zhang, Issam Mudawar, and Mohammad M. Hasan. Experimental and theoretical study of orientation effects on flow boiling CHF. 45(22):4463–4477.

[17] (PDF) A Method for Assessing the Importance of Body Force on Flow Boiling CHF.

[18] George Yadigaroglu and Gad Hetsroni. Nature of Multiphase Flows and Basic Concepts. In George Yadigaroglu and Geoffrey F. Hewitt, editors, *Introduction to Multiphase Flow: Basic Concepts, Applications and Modelling*, pages 1–37. Springer International Publishing.

[19] Thomas J Hanratty. Physics of Gas–Liquid Flows.

[20] Vinesh H. Gada, Mohit P. Tandon, Jebin Elias, Roman Vikulov, and Simon Lo. A large scale interface multi-fluid model for simulating multiphase flows. 44:189–204.

[21] Geoffrey F. Hewitt and George Yadigaroglu. Modelling Strategies and Two-Phase Flow Models. In George Yadigaroglu and Geoffrey F. Hewitt, editors, *Introduction to Multiphase Flow: Basic Concepts, Applications and Modelling*, pages 39–77. Springer International Publishing, 2018.

[22] Wei Shyy, S. S. Thakur, H. Ouyang, J. Liu, and E. Blosch. *Computational Techniques for Complex Transport Phenomena*. Cambridge University Press, 1 edition.

[23] Rosemary M Sugrue. A Robust Momentum Closure Approach for Multiphase Computational Fluid Dynamics Applications.

[24] Eckhard Krepper, Roland Rzehak, and Dirk Lucas. Validation of a closure model framework for turbulent bubbly two-phase flow in different flow situations. 340:388–404.

[25] Rosemary Sugrue, Ben Magolan, Nazar Lubchenko, and Emilio Baglietto. Assessment of a simplified set of momentum closure relations for low volume fraction regimes in STAR-CCM+ and OpenFOAM. 110:79–87.

[26] Yixiang Liao, Tian Ma, Liu Liu, Thomas Ziegenhein, Eckhard Krepper, and Dirk Lucas. Eulerian modelling of turbulent bubbly flow based on a baseline closure concept. 337:450–459.

[27] M Colombo, R Rzehak, M Fairweather, Y Liao, and D Lucas. Benchmarking of CFD Modelling Closures for Two-Phase Turbulent Bubbly Flows.

[28] Sibel Tas-Koehler, Yixiang Liao, and Uwe Hampel. A critical analysis of drag force modelling for disperse gas-liquid flow in a pipe with an obstacle. 246:117007.

[29] Yixiang Liao, Dirk Lucas, Eckhard Krepper, and Roland Rzehak. Flashing evaporation under different pressure levels. 265:801–813, 2013.

[30] Shumon Koga and Miroslav Krstic. Phase change model: Stefan problem. In *Materials Phase Change PDE Control & Estimation: From Additive Manufacturing to Polar Ice*, pages 1–13. Springer International Publishing.

[31] Tobias Jonsson. On the one dimensional Stefan problem.

[32] Leonardo Boledi, Benjamin Terschanski, Stefanie Elgeti, and Julia Kowalski. A level-set based space-time finite element approach to the modelling of solidification and melting processes. 457:111047.

[33] Daniele Andreucci. Lecture notes on the Stefan problem.

[34] G Birkhoff. TAYLOR INSTABILITY AND LAMINAR MIXING.

[35] D.H. Sharp. An overview of Rayleigh-Taylor instability. 12(1):3–18.

[36] Grétar Tryggvason. Numerical simulations of the Rayleigh-Taylor instability. 75(2):253–282.

[37] Jason Goulding, Mehrnaz Ayazi, Tamar Shinar, and Craig Schroeder. Simplified conservative discretization of the Cahn-Hilliard-Navier-Stokes equations. 519:113382.

[38] Arash Hamzehloo, Paul Bartholomew, and Sylvain Laizet. Direct numerical simulations of incompressible Rayleigh-Taylor instabilities at low and medium Atwood numbers. 33(5):054114, 2021.

[39] Shahab Mirjalili, Christopher B. Ivey, and Ali Mani. Comparison between the diffuse interface and volume of fluid methods for simulating two-phase flows. 116:221–238, 2019.

[40] Ziyang Huang, Guang Lin, and Arezoo M. Ardekani. Consistent, essentially conservative and balanced-force Phase-Field method to model incompressible two-phase flows. 406:109192, 2020.

[41] Hang Ding, Peter D.M. Spelt, and Chang Shu. Diffuse interface model for incompressible two-phase flows with large density ratios. 226(2):2078–2095, 2007.

[42] Malcolm J. Andrews and Stuart B. Dalziel. Small Atwood number Rayleigh–Taylor experiments. 368(1916):1663–1679.

[43] Feng Chen, Aiguo Xu, Yudong Zhang, Yanbiao Gan, Bingbing Liu, and Shuang Wang. Effects of the initial perturbations on the Rayleigh-Taylor-Kelvin-Helmholtz instability system. 17(3):33505.

[44] Jan Wilhelm Gärtner. Modeling and simulation of flash evaporation of cryogenic liquid jets, 2024.

[45] Ahmad Mansour and Norbert Müller. A review of flash evaporation phenomena and resulting shock waves. 107:146–168, 2019.

[46] D. Saury, S. Harmand, and M. Siroux. Flash evaporation from a water pool: Influence of the liquid height and of the depressurization rate. 44(10):953–965, 2005.

[47] Yixiang Liao and Dirk Lucas. A review on numerical modelling of flashing flow with application to nuclear safety analysis. 182:116002, 2021.

[48] Osamu Miyatake, Itsuo Tanaka, and Noam Lior. A simple universal equation for bubble growth in pure liquids and binary solutions with a nonvolatile solute. 40(7):1577–1584, 1997.

[49] Quang Dang Le, Riccardo Mereu, Giorgio Besagni, Vincenzo Dossena, and Fabio Inzoli. Computational Fluid Dynamics Modeling of Flashing Flow in Convergent-Divergent Nozzle. 140(10):101102, 2018.

[50] Mamoru Ishii and Takashi Hibiki. *Thermo-Fluid Dynamics of Two-Phase Flow*. Springer New York, 2011.

[51] Gretar Tryggvason, Ruben Scardovelli, and Stéphane Zaleski. *Direct Numerical Simulations of Gas-Liquid Multiphase Flows*. Cambridge Univ. Press, 1. publ edition, 2011.

[52] Volume-Averaged Macroscopic Transport Equations. In E. Blosch, H. Ouyang, J. Liu, S. S. Thakur, and Wei Shyy, editors, *Computational Techniques for Complex Transport Phenomena*, pages 231–259. Cambridge University Press, 1997.

[53] S. Mostafa Ghiaasiaan. *Two-Phase Flow, Boiling and Condensation: In Conventional and Miniature Systems*. Cambridge University Press, 2007.

[54] M Wörner. A Compact Introduction to the Numerical Modeling of Multiphase Flows.

[55] George Yadigaroglu. Interfacial Instabilities. In George Yadigaroglu and Geoffrey F. Hewitt, editors, *Introduction to Multiphase Flow: Basic Concepts, Applications and Modelling*, pages 79–94. Springer International Publishing, 2018.

[56] C. Truesdell and K. R. Rajagopal. *An Introduction to the Mechanics of Fluids*. Birkhäuser, 2000.

[57] Michel O. Deville and Thomas B. Gatski. *Mathematical Modeling for Complex Fluids and Flows*. Springer Berlin Heidelberg, 2012.

[58] Maxwell Hutchinson. A DISSERTATION SUBMITTED TO THE FACULTY OF THE DIVISION OF THE PHYSICAL SCIENCES IN CANDIDACY FOR THE DEGREE OF DOCTOR OF PHILOSOPHY.

[59] S. Dartevelle. Comprehensive Approaches to Multiphase Flows in Geophysics - Application to nonisothermal, nonhomogenous, unsteady, large-scale, turbulent dusty clouds I. Hydrodynamic and Thermodynamic RANS and LES Models.

[60] Arindam Banerjee. Rayleigh-Taylor Instability: A Status Review of Experimental Designs and Measurement Diagnostics. 142(12):120801.

[61] Oleg Schilling. Progress on Understanding Rayleigh–Taylor Flow and Mixing Using Synergy Between Simulation, Modeling, and Experiment. 142(12):120802.

[62] Marco Colombo and Michael Fairweather. Influence of multiphase turbulence modelling on interfacial momentum transfer in two-fluid eulerian-eulerian cfd models of bubbly flows. *Chemical Engineering Science*, 195:968–984, 2019.